CE vs Hybrid Loss Comparison on MobileNet, VGG16, VGG19 & Inception

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1. Introduction

1. The project compares Normal Loss (Cross Entropy) and Custom Hybrid Loss for embryo stage classification.
2. Goal: improve prediction by considering class relationships.

2. Cross Entropy Loss

1. Treats all classes as independent.
2. Formula:

$$\mathcal{L}_{CE} = -\sum_{i=1}^{C} y_i \log(p_i)$$

1. C – Number of classes
Total categories in the problem
Example: 15 embryo stages

2. yi – True label
Actual class (one-hot encoded)
Only correct class has value 1, others 0

3. pi – Predicted probability
Output from model after softmax
Shows how confident the model is for each class

4. log(pi) – Log penalty
If prediction is wrong → loss becomes large
If prediction is correct → loss becomes small

Internally:
Applies softmax
Picks correct class probability
Computes –log(p)

3. Hybrid (Custom) Loss

1. Combines Cross Entropy with ordinal loss.
2. Formula:

$$\mathcal{L}_{Hybrid} = \mathcal{L}_{CE} + \lambda \cdot \frac{1}{C} \sum_{i=1}^{C} (CDF_{pred,i} - CDF_{true,i})^2$$

1. LCE – Cross Entropy Part

1. Normal classification loss
2. Ensures model predicts the correct class

2. CDFpred– Predicted cumulative distribution

1. Sum of predicted probabilities up to class iii
2. Captures ordering of classes

3. CDFtrue – True cumulative distribution

1. Ideal cumulative values based on true label
2. Represents correct order

4. (CDFpred−CDFtrue)^2−

Ordinal error

1. Measures how far prediction is from true class
2. Larger gap = bigger penalty

5. λand 1/C

1. λ: controls importance of ordinal loss
2. 1/C: normalizes over all classes

4. Results

FINAL COMPARISON (Normal Loss vs Custom Loss)

MODEL	LOSS TYPE	ACCURACY	F1 SCORE
MOBILENET	NORMAL (CE)	0.2954	0.3082
VGG16	NORMAL (CE)	0.2242	0.2347
VGG19	NORMAL (CE)	0.2246	0.2207
INCEPTION	NORMAL (CE)	0.2754	0.2686
MOBILENET	CUSTOM (HYBRID)	0.2696	0.2879
VGG16	CUSTOM (HYBRID)	0.2200	0.2228
VGG19	CUSTOM (HYBRID)	0.2275	0.2324
INCEPTION	CUSTOM (HYBRID)	0.3067	0.3203

Cross Entropy works better for MobileNet and VGG models
Hybrid Loss improves Inception performance
No single loss is best for all models
Hybrid loss is useful when class order matters, but performance depends on the model